

CSE24031 LAB 7

1. Tower of Hanoi

```
function return_value = Tower_Of_Hanoi(n)
    n = round(n);
    if(n == 1)
        return_value = 1;
    else
        return_value = 2 * Tower_Of_Hanoi(n-1) + 1;
    end
end

for i = 1:10
    if((2^i)-1 == Tower_Of_Hanoi(i))
        fprintf("The values match! for %d (ans = %d)\n",i,Tower_Of_Hanoi(i));
    else
        fprintf("The values DONT MATCH! for %d\n",i);
    end
end
```

```
The values match! for 1 (ans = 1)
The values match! for 2 (ans = 3)
The values match! for 3 (ans = 7)
The values match! for 4 (ans = 15)
The values match! for 5 (ans = 31)
The values match! for 6 (ans = 63)
The values match! for 7 (ans = 127)
The values match! for 8 (ans = 255)
The values match! for 9 (ans = 511)
The values match! for 10 (ans = 1023)
```

2. Rabbit Problem (Fibonacci)

```
function return_value = Rabbit_Problem(n)
    n = round(n);
    if(n < 3)
        return_value = 1;
    else
        return_value = Rabbit_Problem(n-1) + Rabbit_Problem(n-2);
    end
end

for n = 5:7
    element1 = 1/sqrt(5);
    element2 = (1+sqrt(5))/2;
    element3 = (1 - sqrt(5))/2;
    trash = element1*(element2.^n - element3.^n);
    if(round(trash) == Rabbit_Problem(n))
        fprintf("The values match for n = %d ! (value = %d)\n",n,Rabbit_Problem(n));
    else
        disp("THEY DONT MATCH!!!")
    end
end
```

end

The values match for $n = 5$! (value = 5)
The values match for $n = 6$! (value = 8)
The values match for $n = 7$! (value = 13)

3. Code word enumeration

```
function r = a(n)
    n = round(n);
    if(n == 1) r = 9;
    elseif(n > 1)
        r = 8*a(n-1) + 10^(n-1);
    end
end

for i = 1:5
    fprintf("a%d is %d\n",i,a(i));
end
```

```
a1 is 9
a2 is 82
a3 is 756
a4 is 7048
a5 is 66384
```

question 4

```
function r = a2(x)
    if(x == 1) r = 2;
    elseif(x == 2) r = 3;
    else
        r = a2(x-1) + a2(x-2);
    end
end

for i = 1:8
    fprintf("a%d is %d\n",i,a2(i));
end
```

```
a1 is 2
a2 is 3
a3 is 5
a4 is 8
a5 is 13
a6 is 21
a7 is 34
a8 is 55
```

Other questions

1.

```
function r = t1(n)
    if n == 0
```

```

        r = 0;
    elseif n == 1
        r = 5;
    else
        r = 3*t1(n-1) + 4*t1(n-2);
    end
end
t1(4)

```

```

ans =
255

```

2.

```

function r = t2(n)
    if n == 0 || n == 1
        r = 1;
    else
        r = 4 * (t2(n-1) - t2(n-2));
    end
end
t2(4)

```

```

ans =
-16

```

3.

```

function r = t3(n)
    if n == 0
        r = 7;
    elseif n == 1
        r = 16;
    else
        r = 5*t3(n-1) - 6*t3(n-2);
    end
end
t3(4)

```

```

ans =
242

```

4.

```

function r = t4(n)
    if n == 0
        r = 5;
    elseif n == 1
        r = 16;
    else
        r = 7*t4(n-1) - 10*t4(n-2);
    end
end

```

```
t4(4)
```

```
ans =  
1298
```